

POLITECNICO
MILANO 1863
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

System and Transceiver Architectures

System Architectures

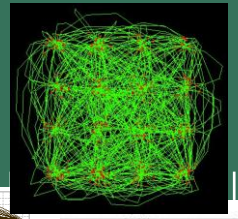
Agenda

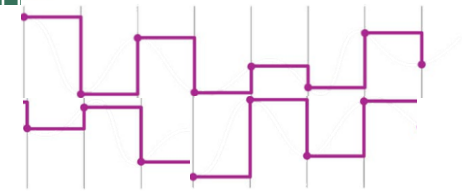
Modern Transceivers: Duplexing

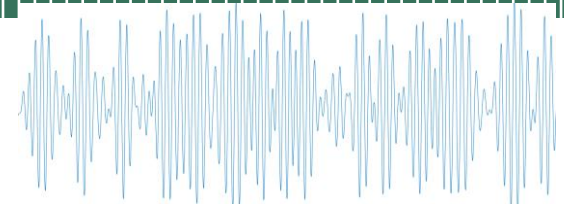
Coordination in Bidirectional Systems

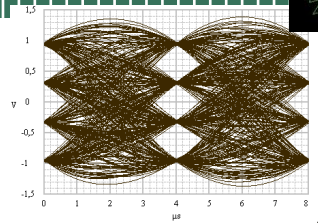
Capacity Increase

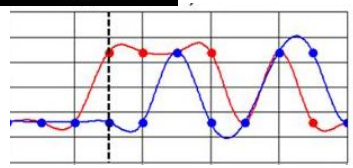
Modern Communication System: as seen so far





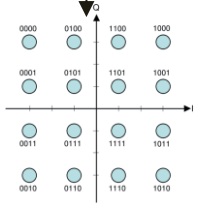


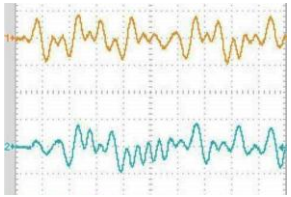


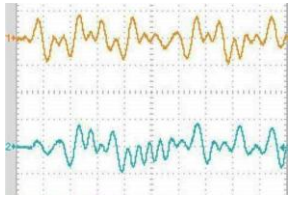


Bits | Mapping to QAM | Symb. | Shaping Filter (RRC) | I&Q | Baseband Proc. | I&Q | IQ Modul, Upconv+ amplifiers | RF | Channel + eq. noise | RF | Amp+ Downcnv +IF filt+ IQDem | I&Q | Baseb. Proc. | I&Q | Shaping Filter (RRC) | Symb. | Sampling | Bits | Decision & De-mapping

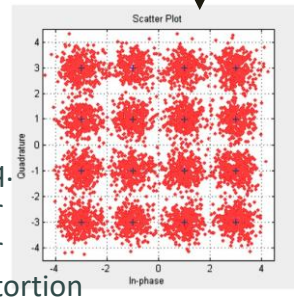
Digital | Analog (or sometimes partly digital) | Analog (or sometimes partly digital) | Digital





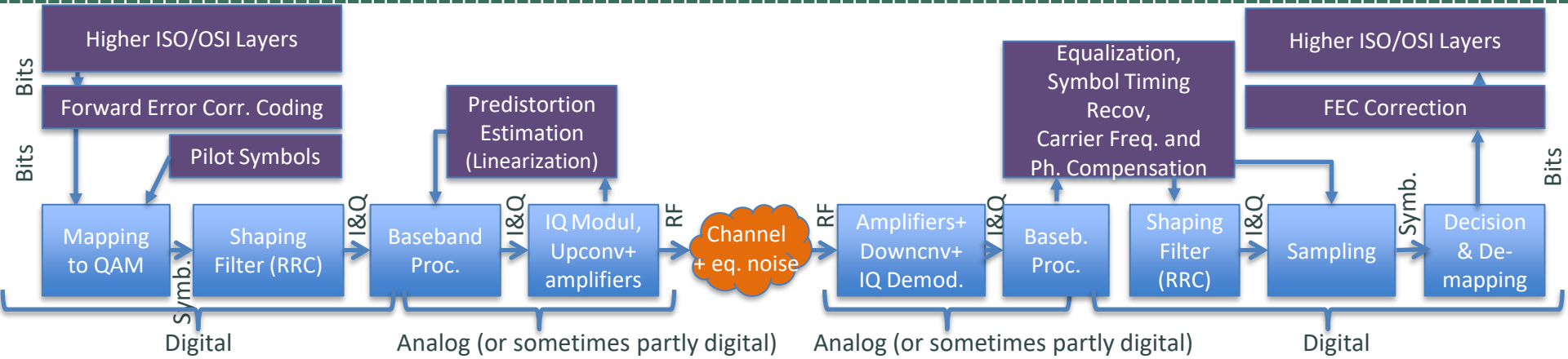


Includes:
Thermal noise
Interferers
Phase and freq. error of carrier
Sampling error
Non-linear distortion



Unidirectional systems: “simplex”, either Tx or Rx
Example: Broadcast TV (DVB-T, Satellite…) and radio

Modern Communication System: Broader View



Forward Error Correction: add overhead to be more resilient to random errors

→ Reed Solomon, LDPC, Trellis, Viterbi…

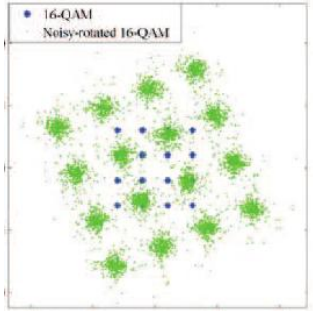
Pilot symbols: known symbols (usually at high power) periodically transmitted

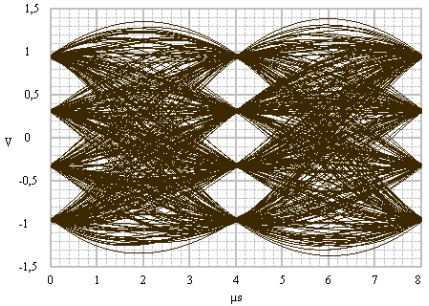
→ Estimate carrier frequency and phase errors and symbol clock

→ Estimate channel response and equalize: compensate some linear distortions (i.e. multipath, imperfect filters…)

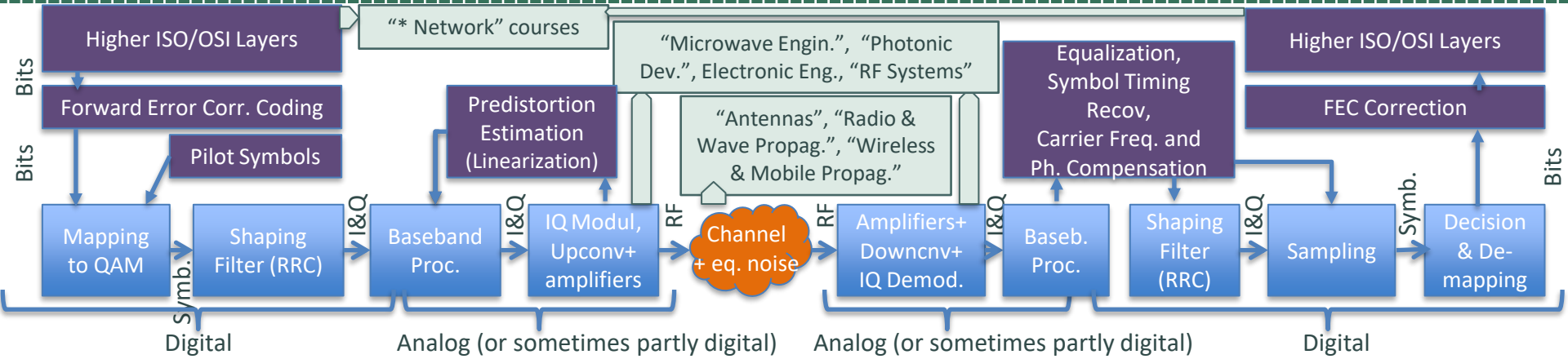
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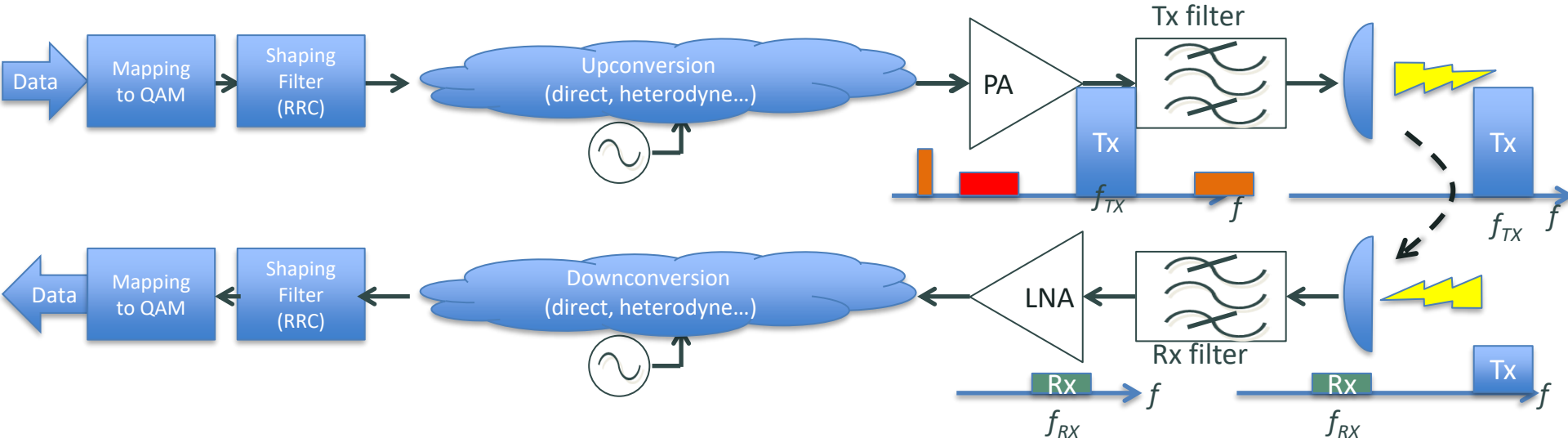
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Course "Information Theory"

Courses "Digital Communications I and II", "Wireless Communications"

Digital Transceiver: Frequency Division Duplexing (FDD)



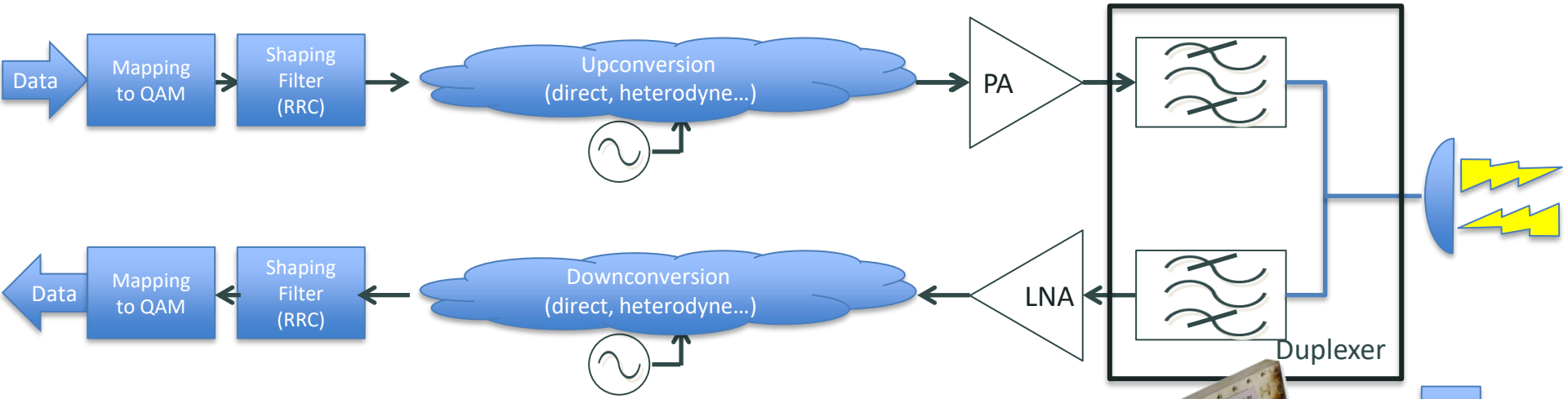
Duplex operation: Tx and Rx at the same time

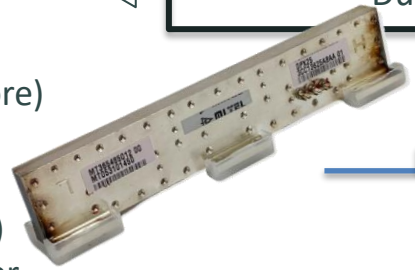
Tx filter: limit out-of-band emissions & spurious signals in Rx band (not filtered by Rx filter)

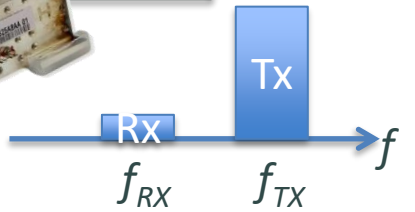
Rx filter: image filter and reject high-power Tx signals (not filtered by Tx filter, saturating LNA)

This requires two antennas! When antenna is “large” (i.e. only one fits on an equipment)→…

Digital Transceiver: Frequency Division Duplexing (FDD)

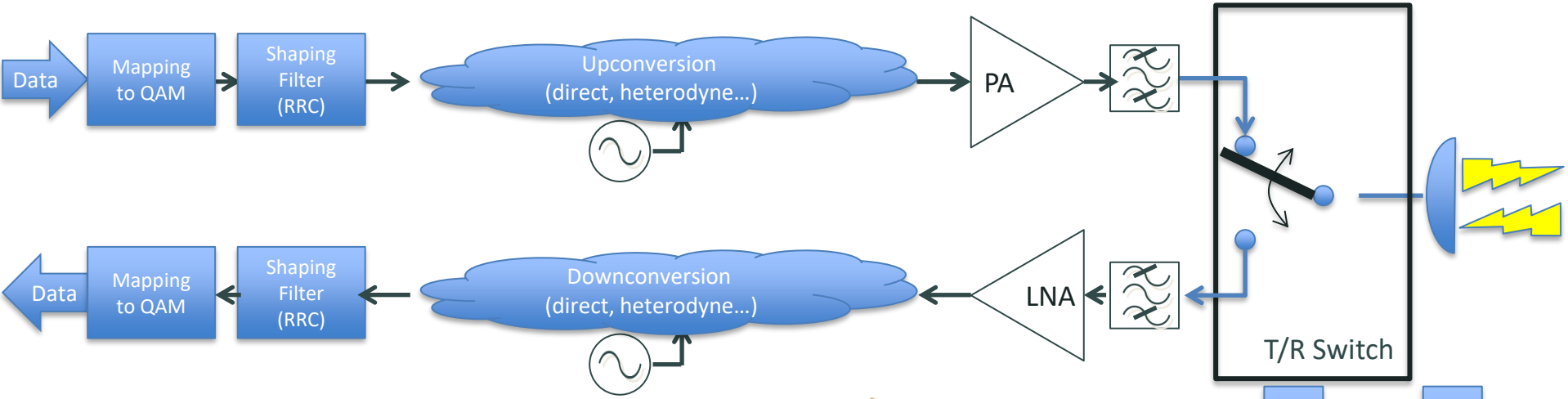


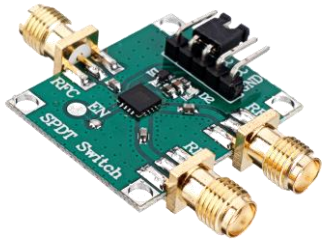


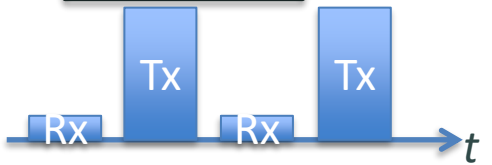


Duplexer: allows FDD operation; similar to two filters (same purpose as before)
Often expensive (very low loss, high performance)
Fixed duplex spacing: $|f_{Tx}-f_{RX}|$
The two ends require slightly different hardware (at least reversed duplexer)
Variation: "Flexible FDD", with variable duplex spacing → Limited by duplexer

Digital Transceiver: Time Division Duplexing (TDD)

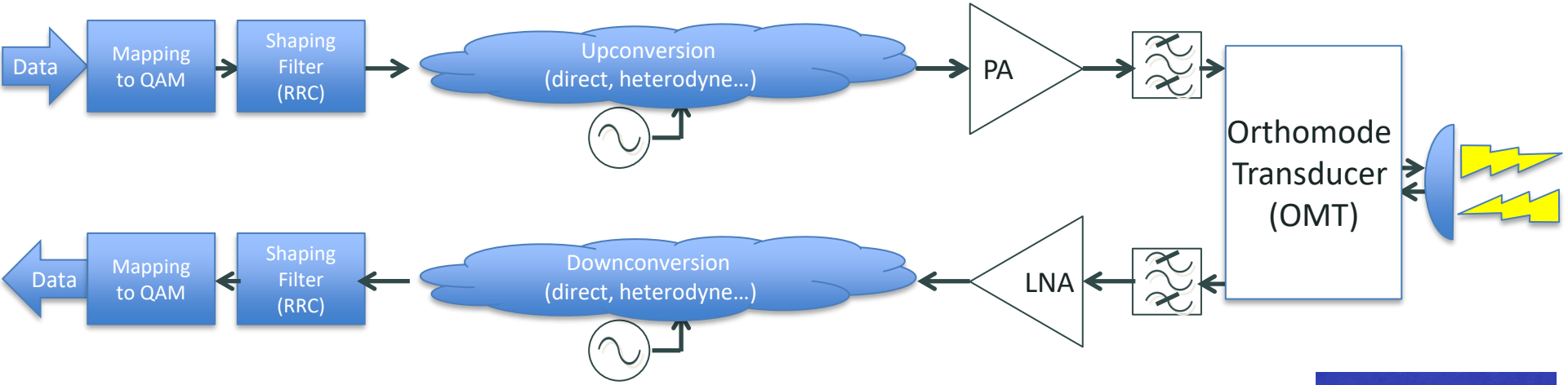






Switch: allows TDD: time slot for Tx, time slot for Rx
Cheaper; more flexible for asymmetric traffic
Requires synchronization between the two ends
Slight penalty due to guard intervals (time)
What is the total bandwidth?

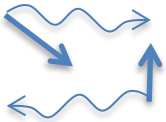
Digital Transceiver: Polarization Duplexing



Polarization duplexing: Tx on one polarization (i.e. horizontal)
Rx on the orthogonal one (i.e. vertical)

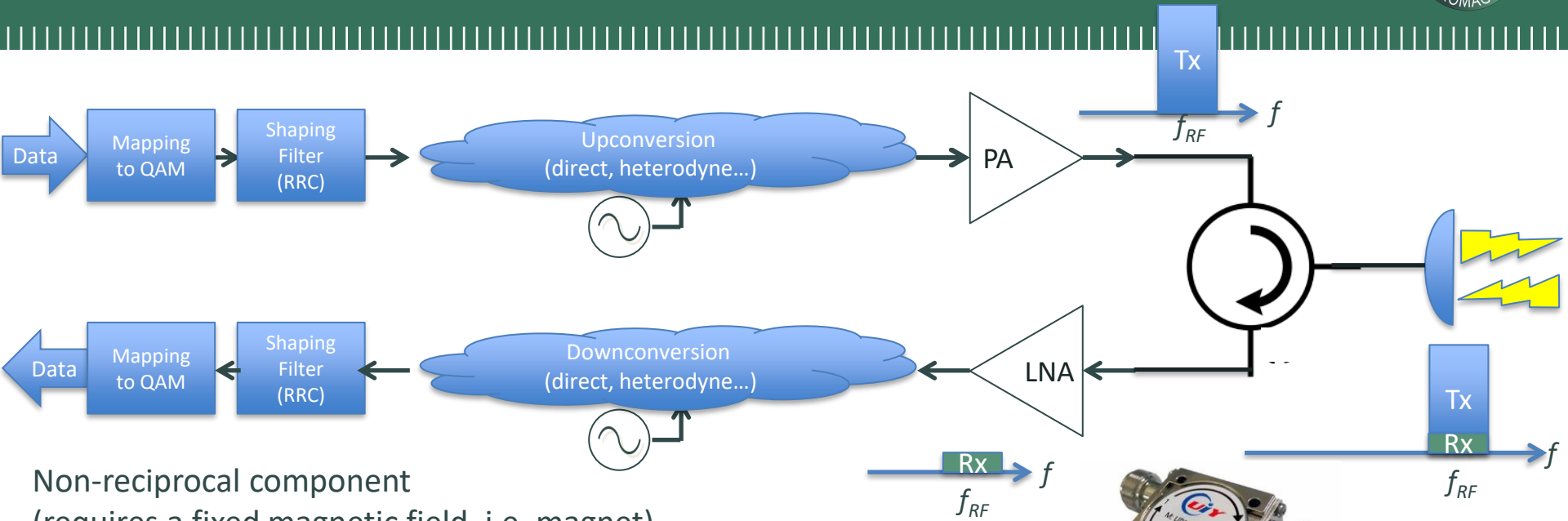
Requires a dual-polarization antenna

Used rarely





Digital Transceiver: Circulator



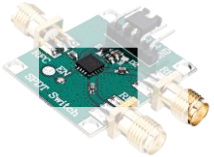


Non-reciprocal component
(requires a fixed magnetic field, i.e. magnet)
For instance: clockwise propag., counterclockwise isol.
True full duplex! Used in radars
Very sensitive to reflections

Components

Components for Duplexing

Switch



Solid state, MMICs (not mechanical, too slow)

Diplexer



Often bulky, cavities, distributed resonators

Orthomode Transducer



Often bulky, in waveguide technology

Circulators

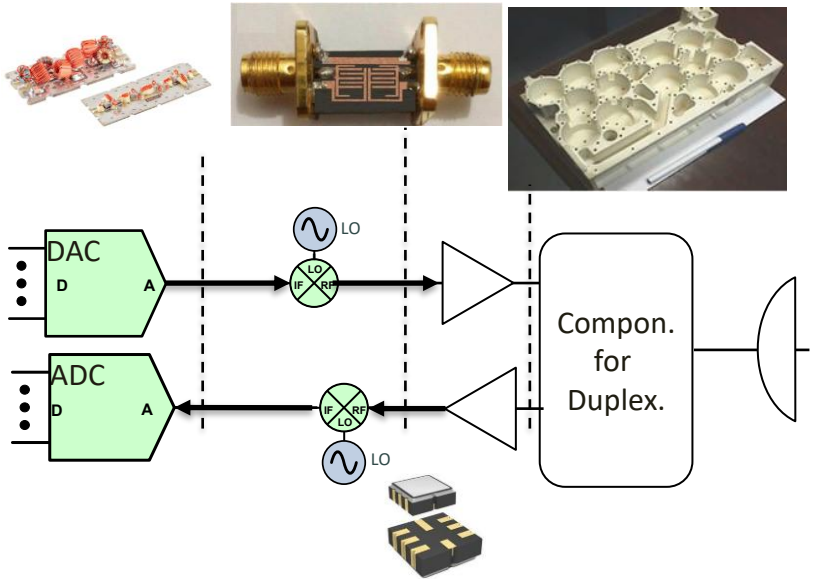


Often bulky, with ferrites and permanent magnets

Components: Filters

Filters are everywhere!

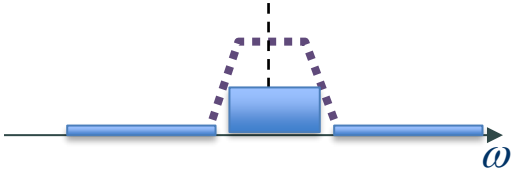
- After DAC: suppress digital noise & spurious
- Before ADC: avoid aliasing
- After Downconv mixer: reject undesired mixing products
- After Upconv mixers: reject undesired mixing products
- Before Downconv mixer: reject image band
- After Oscillator: suppress harmonics & limit ph. noise
- After Power Amplifier: suppress out-of-band emissions
- Before LNA: limit out-of-band noise and interf.



Nomenclature often found in RF Systems

Attenuation vs Rejection

- "Rejection": a signal does not pass through a component (i.e. a filter) when the component reflects the signal back at the input
- "Attenuation": reduced transmission due to losses (dissipation)
  → Also called **Insertion Loss (IL), normally >0dB (e.g. 0.8dB)**



A bandpass filter exhibits some attenuation in its passband

→ Adds noise and the input noise also passes through

In its stopband transmission is very small but the signal is normally back-reflected

→ Does not add noise in its stopband but may allow a tiny portion of input noise through

Return Loss: Portion of signal in the passband which is reflected due to impedance mismatches (more on this after introducing S parameters): normally >0 in dB (e.g., 15dB, 20dB). Finite RL implies some Insertion Loss

Crosspolar Discrimination (XPD): Fraction of a signal in a polarization which leaks to the other (e.g. XPD=-10Log(PXpol/PTotal), normally positive in dB (e.g. XPD=30dB))

Transmission Zeros: Specific frequencies at which a component (e.g., filter) does not transmit from one port to the other

Order of filter: Number of resonators or cavities or "poles" of the response

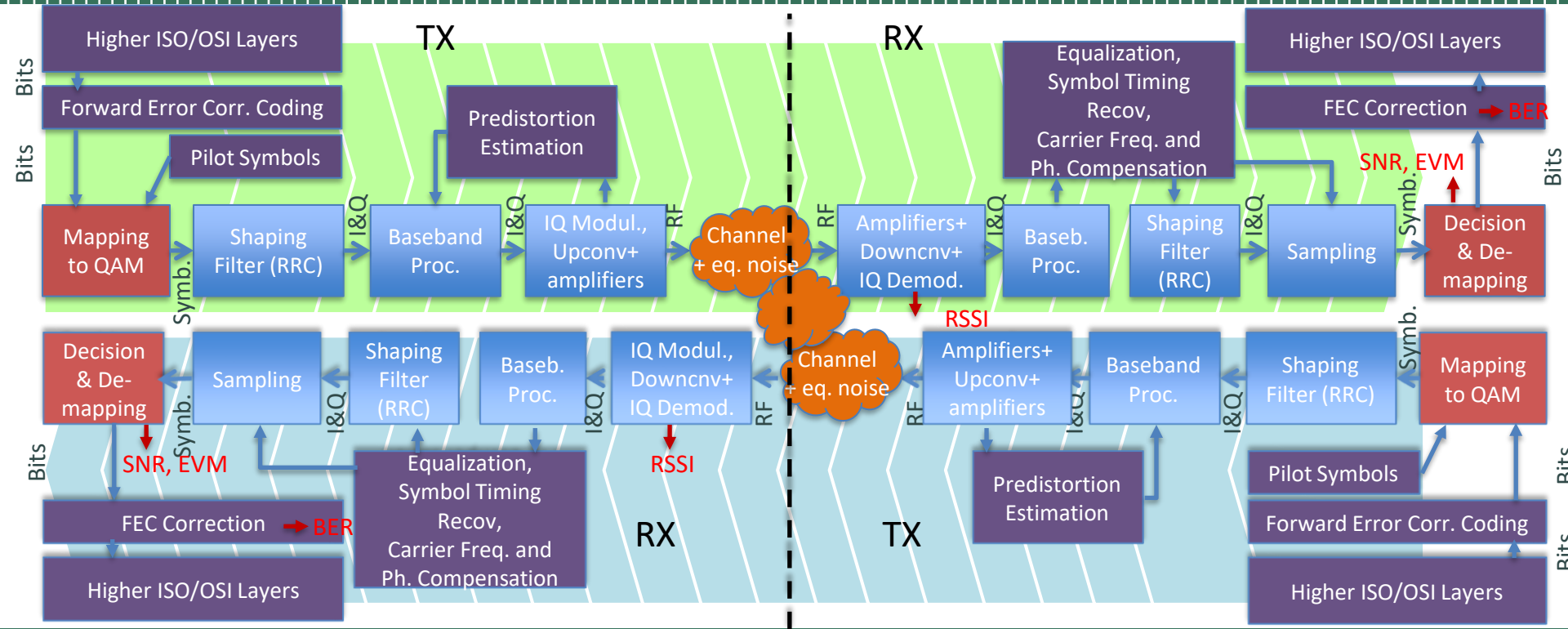
Agenda

Modern Transceivers: Duplexing

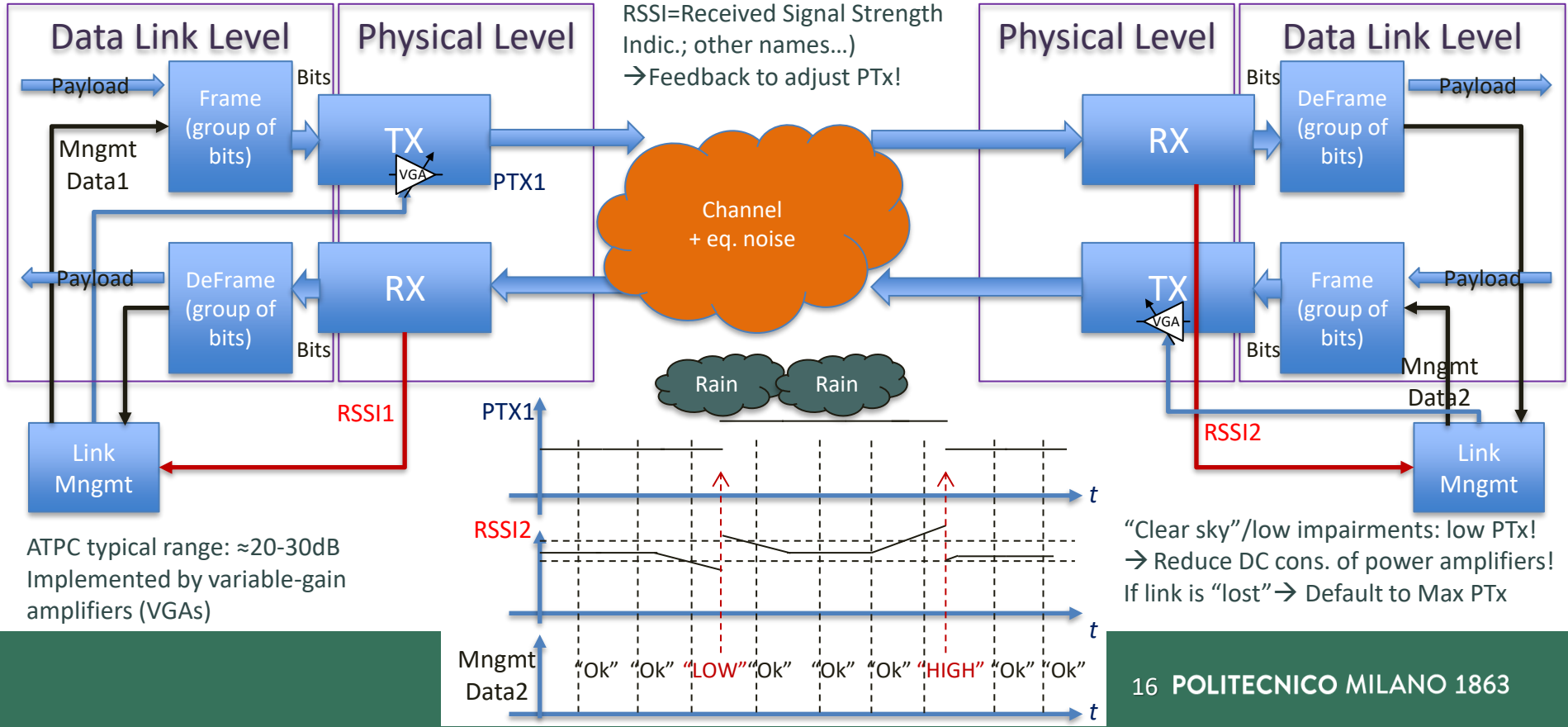
Coordination in Bidirectional Systems

Capacity Increase

Coordination between Two Ends



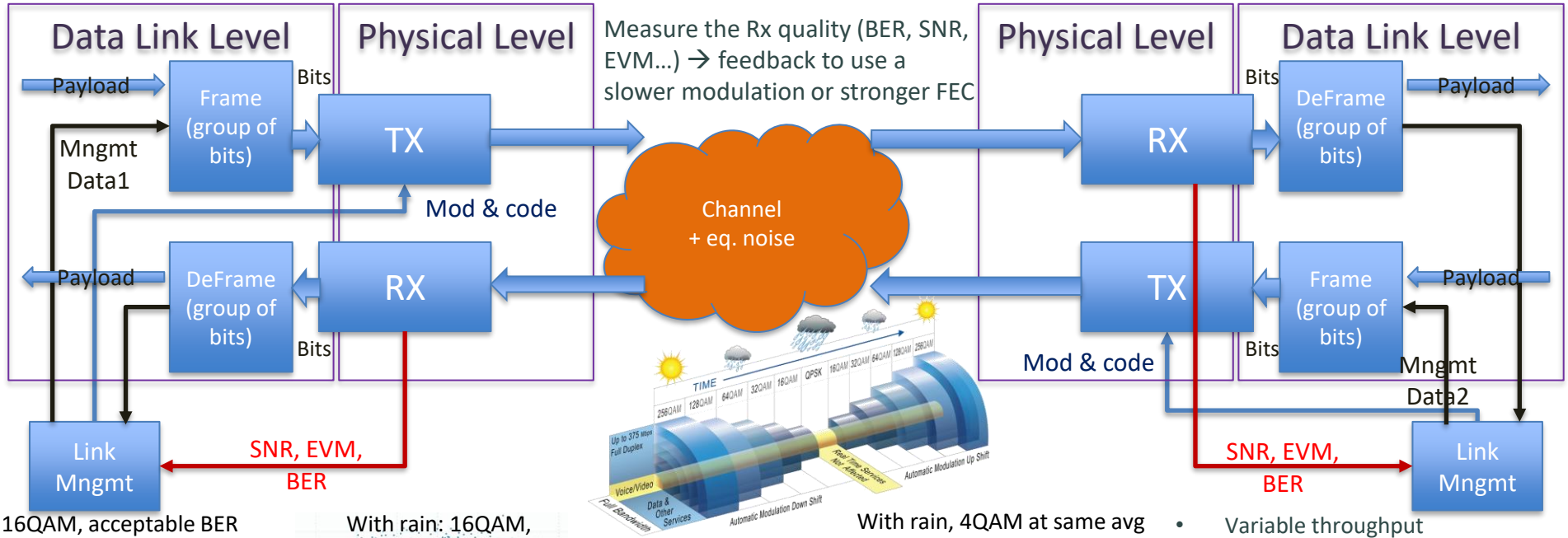
Coordination: Automatic Tx Power Control in Action



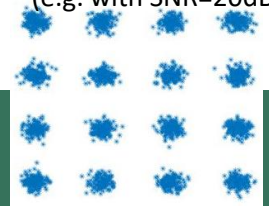
ATPC typical range: ≈20-30dB
Implemented by variable-gain amplifiers (VGAs)

"Clear sky"/low impairments: low PTx!
→ Reduce DC cons. of power amplifiers!
If link is "lost"→ Default to Max PTx

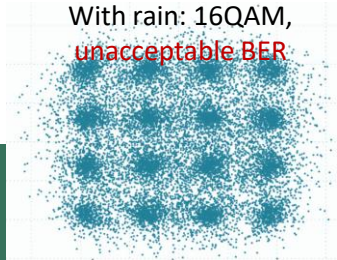
Coordination: Adaptive Coding & Modulation



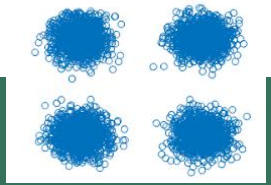
16QAM, acceptable BER (e.g. with SNR=20dB)



With rain: 16QAM, unacceptable BER



With rain, 4QAM at same avg power, now acceptable BER



- Variable throughput
- If one direction of the link is lost →default to most robust modulation & FEC

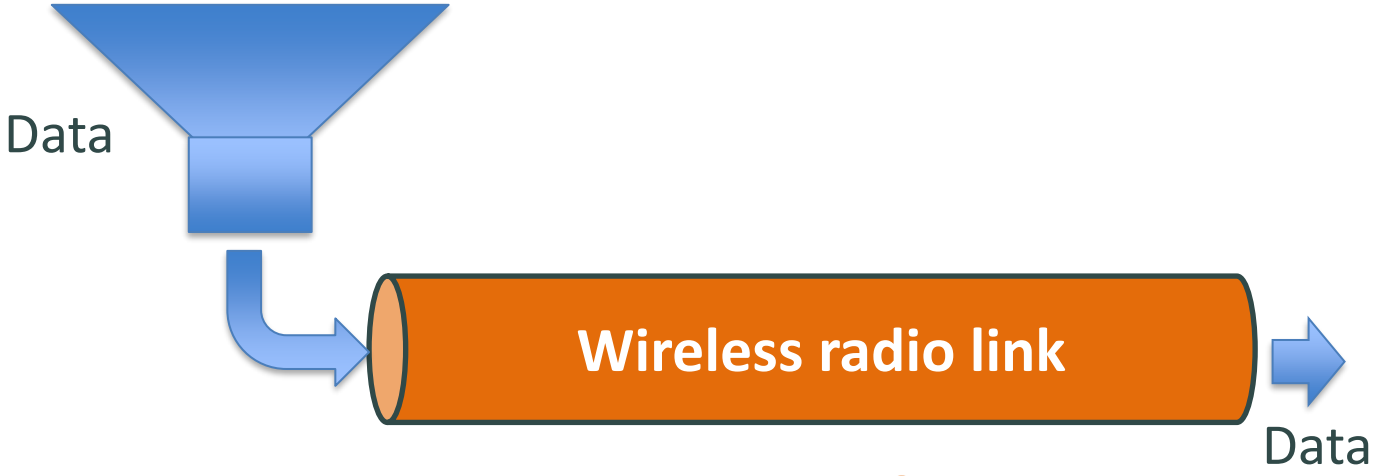
Agenda

Modern Transceivers: Duplexing

Coordination in Bidirectional Systems

Capacity Increase

Capacity Enhancement



Total data rate (bit/sec) → $R = S \cdot D \cdot N$

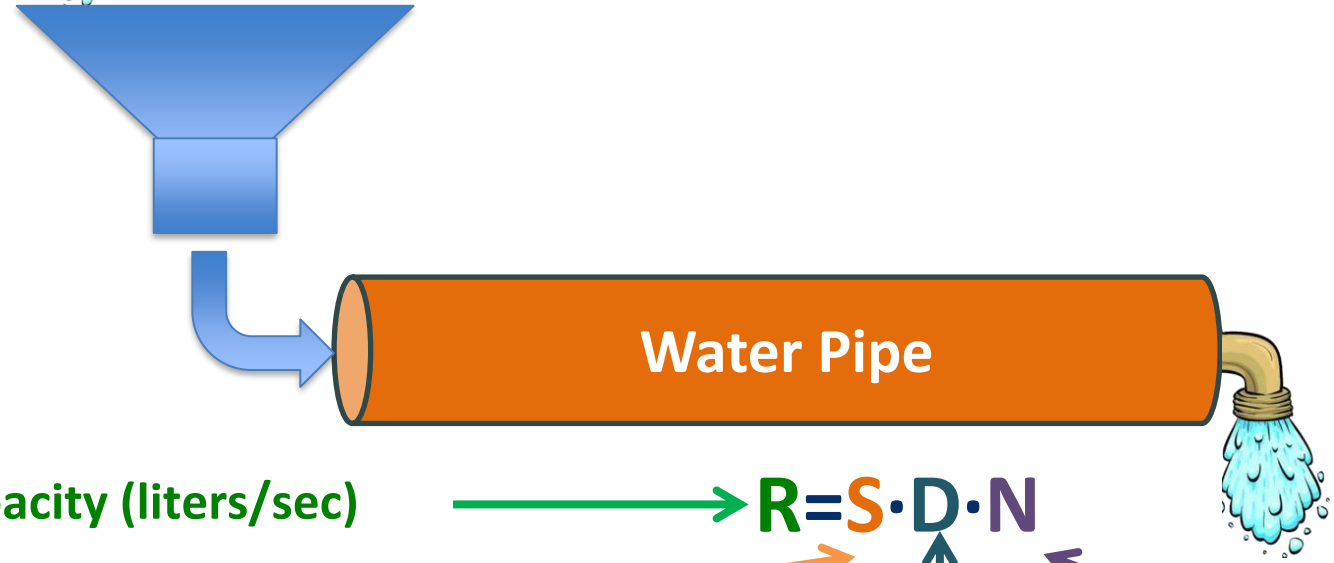
Symbol rate (symbols/sec): $S = B/(1+\alpha)$

Modulation (bits/symbol) $D = \log_2(M)$ [e.g. M-QAM]

Number of links

Capacity Enhancement: Microwave Plumbers





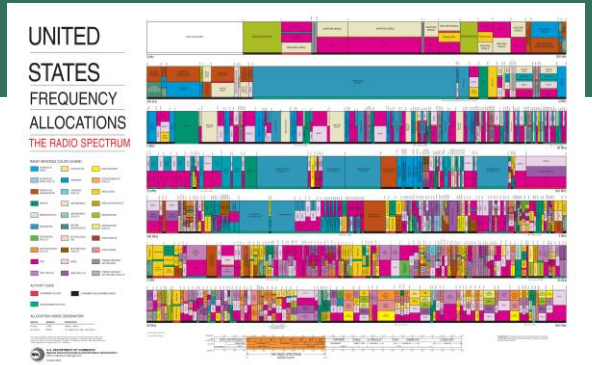
Capacity (liters/sec) → $R = S \cdot D \cdot N$

Pipe section (m^2)

Flow velocity (→pressure)

Number of pipes

Capacity Enhancement: “Pipe Section”



Data

Wireless radio link

Data

Increasing S:
Roll-off α helps, but only a little (higher latency)
→Larger bandwidth B
→Expensive
→Critical RF components



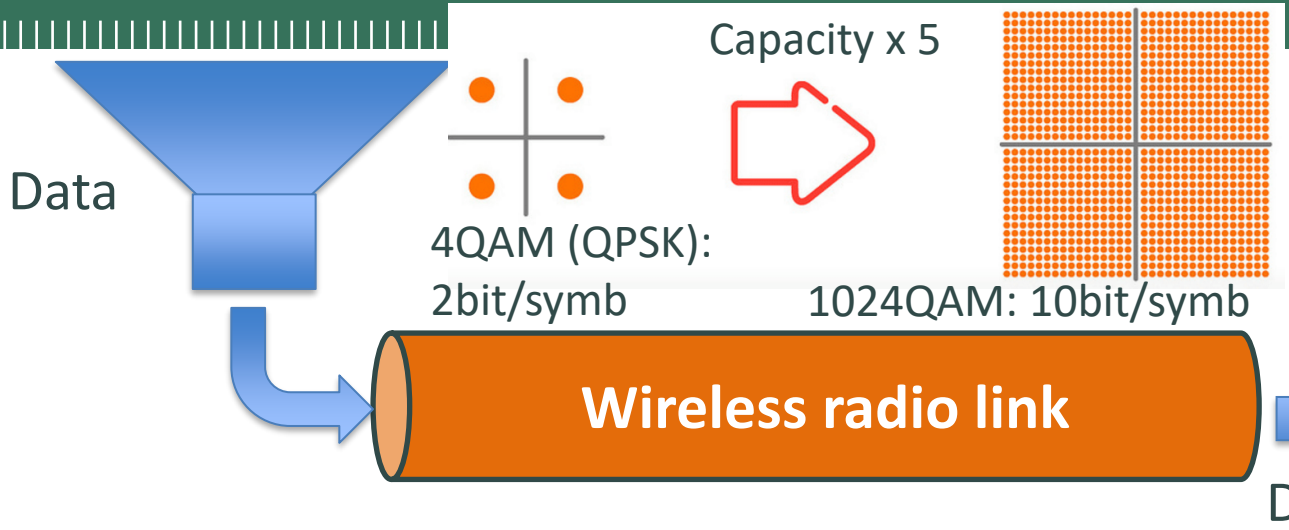
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Symbol rate (symbols/sec):
$S = B/(1+\alpha)$

Modulation (bits/symbol)
$D = \log_2(M)$ [e.g. M-QAM]

Number of links

Capacity Enhancement: “Pipe Pressure”



Increasing D:
Requires a larger Eb/N0, thus a larger SNR to preserve BER
→More Tx power
→More directive antennas
→Less noise (close to technological limits)

Total data rate (bit/sec) → $R = S \cdot D \cdot N$

Symbol rate (symbols/sec): $S = B/(1+\alpha)$

Modulation (bits/symbol) $D = \log_2(M)$ [e.g. M-QAM]

Number of links

Capacity Enhancement: “More Pipes”



Increasing N:
More channels (frequency or time slots)→ Expensive
Or….?

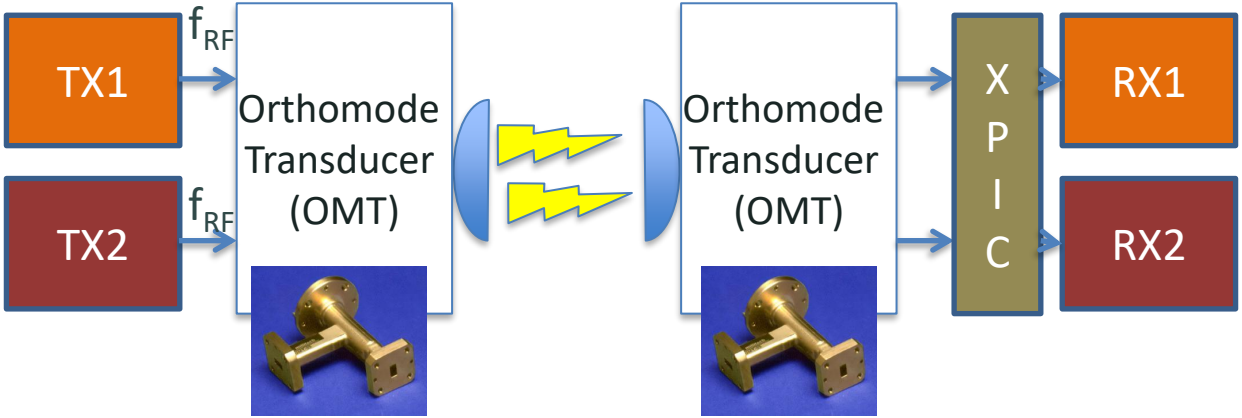
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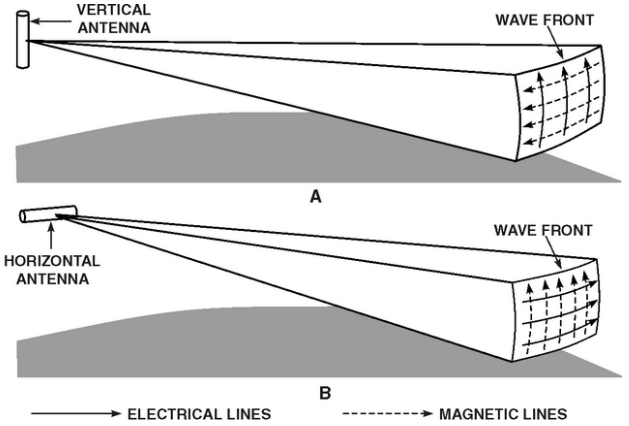
Symbol rate (symbols/sec): $S = B/(1+\alpha)$

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Number of links

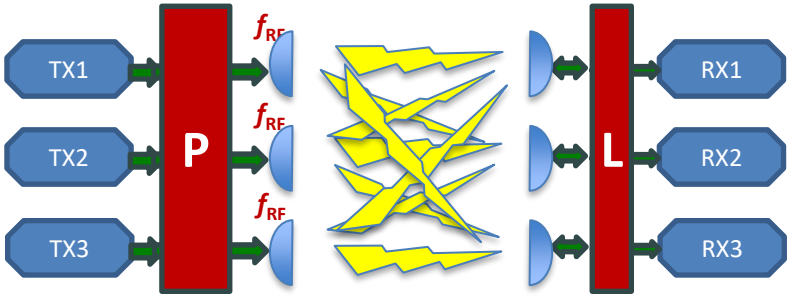
Capacity Enhancement: “More Pipes”: Dual polarization

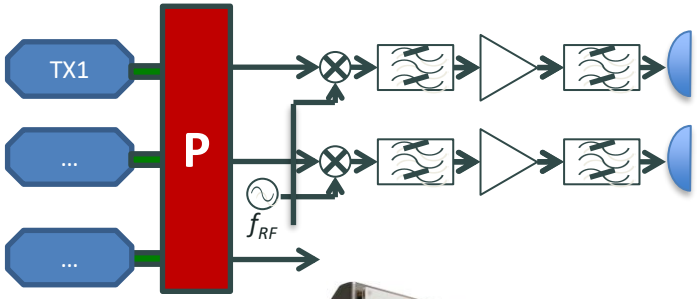




Requires Orthomode Transducers and Dual-Polarization antennas
Exploiting orthogonality of the two radiated waves
Some interference at RX is anyway observed (15-25dB) due to tolerances, propagation…
Adaptive cancelling required (usually digital, Cross Polarization Interference Cancelling, XPIC)
Cheapest way to increase capacity! But only two-fold (no more orthogonal polarizations available)

Capacity Enhancement: “More Pipes”: MIMO



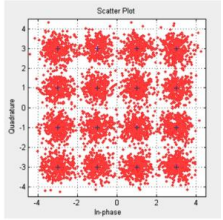


- Complete interference at RX!
- But: interfering signals (at same carrier frequency) may be separated
- Needs several Tx and Rx antennas available
- Needs physical spacing (spatial diversity)
  or different propagation paths (i.e. bounces on obstacles, path diversity)
- Distance: “far field” of each antenna, but “near field” of overall antenna system
- Processing is entirely done in digital domain
- Several Tx and Rx chains
- Preprocessing possible (SVD)



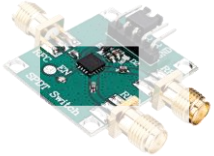
4G and 5G use MIMO and massive-MIMO (mMIMO)

Wrap-Up









Review of Unidirectional systems

- Fundamental cooperation of analog and digital stages

Bidirectional systems (transceivers)

- FDD, TDD… and components (duplexer, switch, OMT, circulator, filters)
- not to be confused with TDMA, FDMA, CDMA…
  (these are schemes for multiple access)
- Cooperation between the two ends: ATPC, ACM

Capacity increase

- Symbol rate (bandwidth-limited);
- modulation schemes (noise&distortion-limited);
- Dual polarization (x2 only); MIMO (limited by antenna spacing)

